ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MATHEMATICAL LITERACY

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) fundamental subject, Level 3. Finance – banking, exchange rates, inflation, break-even and income tax (35), measurement (25), maps and travel (10), data handling (20), probability (10). All people, prices and data are fictional and were created for this examination.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number the answers according to the numbering system used in this question paper.
3. Show ALL calculations. Answers only will not necessarily earn full marks.
4. An approved calculator (non-programmable and non-graphical) may be used.
5. Round off answers to TWO decimal places, unless stated otherwise. Write units with all answers.
6. Write neatly and legibly.

QUESTION 1: FINANCE

1.1 Thandi's bank charges the following fees: monthly account fee R65,00; cash withdrawal at the bank's own ATM R8,50 plus R1,20 per R100 (or part of R100) withdrawn; debit order R4,50 each. In one month she withdraws R1 500 twice and has three debit orders. Calculate her total bank fees for the month. (5)

1.2 Sipho travels to the United Kingdom. The exchange rate is £1 = R23,40, and the bank charges 1,5% commission on the rand amount.

1.2.1 How many rand does he need to buy £850, before commission? (2)

1.2.2 Calculate the total cost including commission. (2)

1.3 A car costs R180 000 today. If its price increases with inflation at 5,5% per year, what will it cost in 3 years' time? (3)

1.4 Mama Zodwa sells vetkoek. Her fixed costs (stall rent and gas) are R1 800 per month. Each vetkoek costs R3,50 to make and sells for R8,00.

1.4.1 Calculate the number of vetkoek she must sell per month to break even. (3)

1.4.2 Calculate her profit in a month when she sells 650 vetkoek. (3)

1.4.3 Draw the graphs of total income and total cost on the same set of axes for 0 to 600 vetkoek, and mark the break-even point. (4)

1.4.4 Explain what the break-even point means for her business. (2)

1.5 Use the simplified tax table below to calculate the monthly income tax of a worker whose taxable income is R240 000 per year.

Taxable income (R per year)	Rate of tax
1 – 237 100	18% of taxable income
237 101 – 370 500	42 678 + 26% of taxable income above 237 100
Primary rebate (under 65): R17 235 per year	

(6)

1.6 Two cellphone contracts are available. Contract A: R199 per month including 150 free minutes, then R1,50 per extra minute. Contract B: R59 per month plus R1,20 per minute.

1.6.1 Calculate the cost of each contract for a person who talks for 100 minutes per month. (2)

1.6.2 Calculate the cost of each contract for 150 minutes per month. (2)

1.6.3 Advise a student who talks for about 100 minutes per month which contract to choose. (1)

[35]

QUESTION 2: MEASUREMENT

2.1 A house has a rectangular roof area of 10 m by 8 m (measured flat). Rainwater is collected from the roof into a cylindrical tank with a radius of 0,8 m and a height of 1,6 m. Use $\pi = 3{,}14$.

2.1.1 Calculate the volume of rainwater (in litres) collected from 25 mm of rain. (1 m^3 = 1 000 ℓ) (4)

2.1.2 Calculate the capacity of the tank in litres. (3)

2.1.3 Will the tank hold all the water from 25 mm of rain? Explain. (1)

2.2 A concrete slab measuring 4 m by 3 m must be 100 mm thick. Seven bags of cement are needed per cubic metre of concrete, and one bag costs R105.

2.2.1 Calculate the volume of the slab in m^3. (3)

2.2.2 Calculate the number of bags of cement needed and the cost. (3)

2.3 A car uses 7,5 ℓ of petrol per 100 km. Petrol costs R23,45 per litre. Calculate the petrol cost of a 640 km trip. (5)

2.4 A nurse must give a child 15 mg of medicine per kg of body mass. The child has a mass of 18 kg. The syrup contains 120 mg in every 5 mℓ.

2.4.1 Calculate the dose in mg. (2)

2.4.2 Calculate the volume of syrup in mℓ that must be given. (4)

[25]

QUESTION 3: MAPS AND TRAVEL

3.1 A bus travels 145 km from Mthatha to East London at an average speed of 90 km/h.

3.1.1 Calculate the travelling time in hours and minutes. (3)

3.1.2 If the bus leaves at 07:15, at what time will it arrive? (2)

3.2 On a map, 4 cm represents 10 km. Write this as a number scale (1 : …). (3)

3.3 Give TWO reasons why the actual travel time may be longer than the calculated time. (2)

[10]

QUESTION 4: DATA HANDLING

4.1 The monthly electricity use (in kWh) of a household over one year is: 320 280 350 410 460 520 540 500 430 380 330 300 (January to December)

4.1.1 Arrange the data in ascending order. (1)

4.1.2 Calculate the mean monthly use. (2)

4.1.3 Determine the median, the lower quartile and the upper quartile. (4)

4.1.4 Calculate the range and the interquartile range. (2)

4.1.5 Draw a box-and-whisker plot of the data. (4)

4.1.6 In which months is electricity use the highest? Suggest a reason. (2)

4.2 A company shows its sales on a bar graph whose vertical axis starts at R900 000 instead of R0. Explain how this can mislead readers. (2)

4.3 Explain the difference between a sample and a population, using an example. (3)

[20]

QUESTION 5: PROBABILITY

5.1 A box contains 10 light bulbs, of which 3 are faulty. Two bulbs are taken out one after the other without replacement.

5.1.1 Draw a tree diagram to show all the outcomes and their probabilities. (4)

5.1.2 Calculate the probability that both bulbs are faulty. (2)

5.1.3 Calculate the probability that at least one bulb is working. (2)

5.2 A taxi arrived late on 18 of the last 60 days. Use this to estimate the probability that it will be late tomorrow. (2)

[10]

TOTAL: 100

END OF QUESTION PAPER